# 历史记录

V1.0 版本 发布

V1.1 修改Power Key设计电路；具体请参考附件电路参数调整

V1.2 修改BT天线网络电路，调整蓝牙天线规格

修改Corevdd C12 电容 0.1uF改为1uF

修改TF卡检测电阻R31 2.2K改为1K

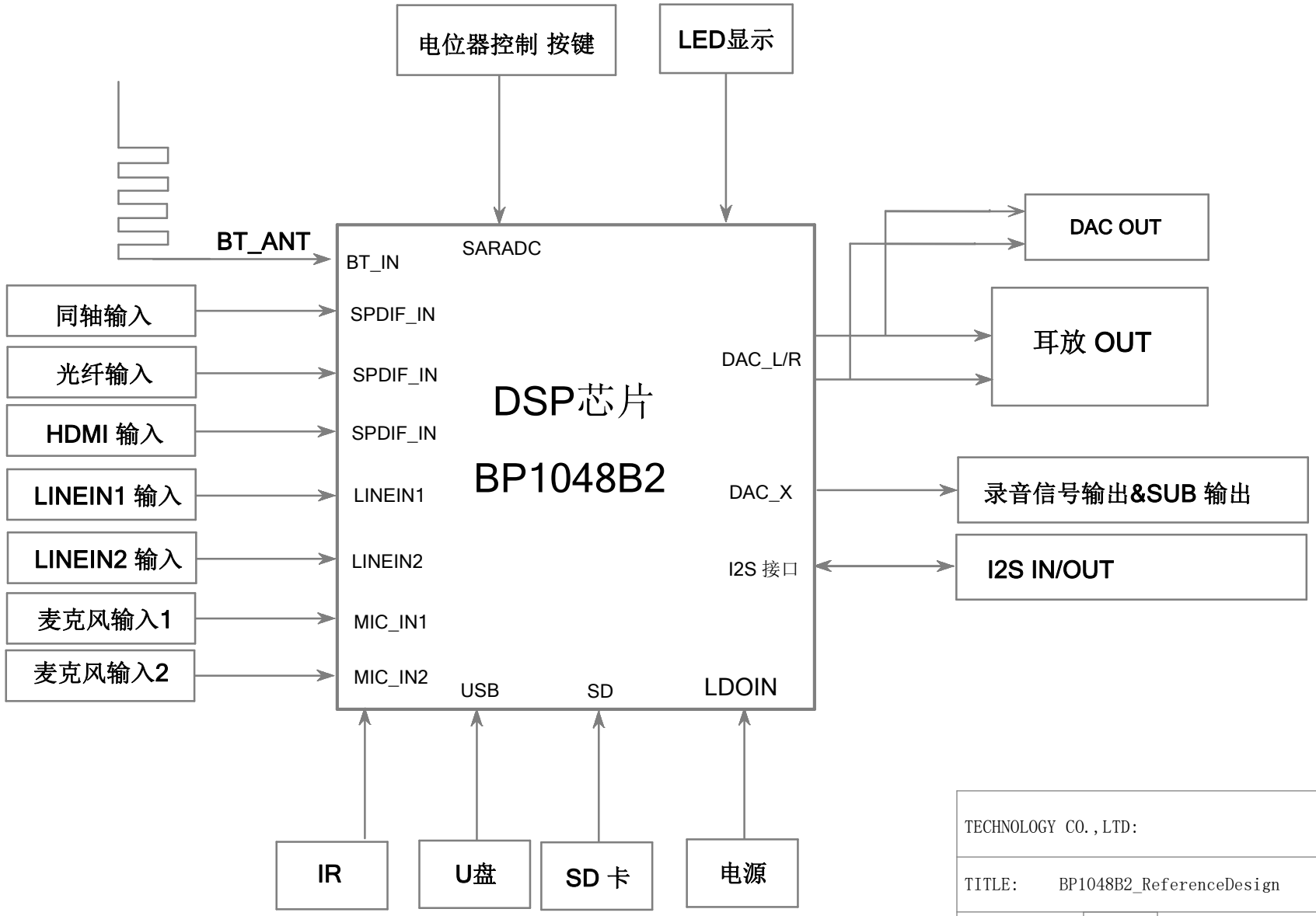
增加光纤/同轴/HDMI 20pF下地电容，改善noise

**HDMI CEC口只能配置复用在GP_A0/A8/A22 TIMER3功能**

Power Key 管脚上需要增加C51 104电容；

修改Powerkey Key电阻R51位置

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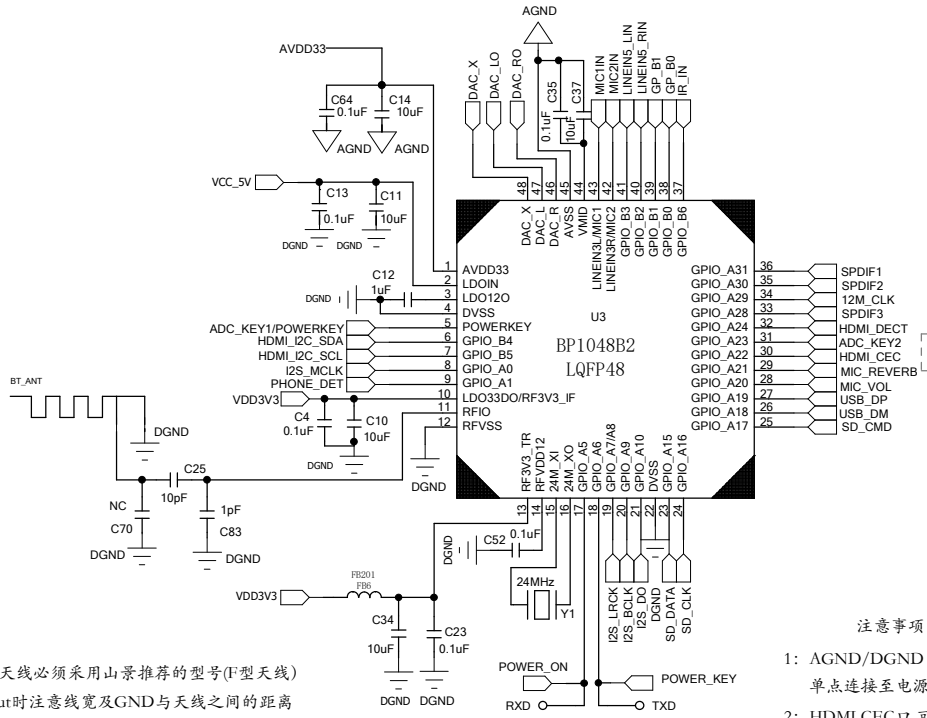


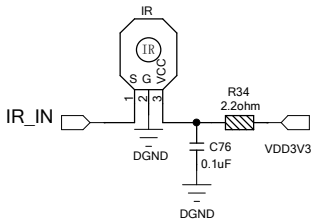

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# BP1048B2 IO基本信息

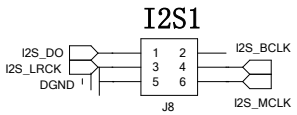
| Pin Name | Pin No | Other Function Assignment |
|---|---|---|
| GPIO_A[0] | 8 | UART0_RXD / UART0_TXD / MCLK0_OUT / MCLK0_IN /TIM3_PWM |
| GPIO_A[1] | 9 | UART0_TXD / UART0_RXD / TIM4_PWM |
| GPIO_A[5] | 17 | LCD0/SPIM_MOSI / UART0_RXD / I2C_SDA |
| GPIO_A[6] | 18 | LCD1/SPIM_CLK / UART0_TXD / I2C_SCL |
| GPIO_A[7/8] | 19 | LCD2/SPIM_MISO / UART0_CTS / MCLK1_OUT /MCLK1_IN/ UART0_RTS / I2S1_LRCLK / TIM3_PWM |
| GPIO_A[9] | 20 | LCD4/UART1_RXD / I2S1_BCLK / TIM4_PWM |
| GPIO_A[10] | 21 | LCD5/UART1_TXD / I2S1_DO / I2S1_DI / TIM5_PWM |
| GPIO_A[15] | 23 | LCD10/SD_DAT |
| GPIO_A[16] | 24 | LCD11/SD_CLK |
| GPIO_A[17] | 25 | LCD12/SD_CMD |
| GPIO_A[18] | 26 | USB_DM / UART1_RXD |
| GPIO_A[19] | 27 | USB_DP / UART1_TXD |
| GPIO_A[20] | 28 | AD0 / SD_DAT / SPIM_MOSI / SPIS_MOSI / I2S0_LRCLK /I2S1_LRCLK |
| GPIO_A[21] | 29 | AD1 / SD_CLK / SPIM_CLK / SPIS_CLK / I2S0_BCLK / I2S1_BCLK |
| GPIO_A[22] | 30 | AD2 / SD_CMD / SPIM_MISO / SPIS_MISO / I2S0_DO / I2S0_DI /TIM3_PWM |
| GPIO_A[23] | 31 | LCD13/AD0 / SPIS_CS/ I2S0_DI / I2S0_DO / TIM4_PWM |
| GPIO_A[24] | 32 | LCD14/AD1 / MCLK0_OUT/MCLK0_IN / TIM5_PWM |
| GPIO_A[28] | 33 | AD5 / SPDIF_AI_0 / I2S1_LRCK / SPDIF_DI / SPDIF_DO /TIM4_PWM |
| GPIO_A[29] | 34 | AD6 / SPDIF_AI_1 / I2S1_BCLK / SPDIF_DI / SPDIF_DO / CLK_OUT/IR |
| GPIO_A[30] | 35 | AD7 / SPDIF_AI_2 / I2C_SDA / SPDIF_DI / SPDIF_DO / DMIC1_DAT |
| GPIO_A[31] | 36 | AD8 / SPDIF_AI_3 / I2C_SCL / I2S1_DI / I2S1_DO / SPDIF_DI /SPDIF_DO / DMIC1_CLK |
| GPIO_B[0] | 38 | LINEIN4_R / AD9 / TIM5_PWM / SW_CLK |
| GPIO_B[1] | 39 | LINEIN4_L / AD10 / TIM6_PWM / SW_D |
| GPIO_B[2] | 40 | LINEIN5_R / DMIC0_DAT |
| GPIO_B[3] | 41 | LINEIN5_L / DMIC0_CLK |
| GPIO_B[4] | 8 | I2C_SDA |
| GPIO_B[5] | 9 | I2C_SCL |
| GPIO_B[6] | 37 | EFUSE VDD / CLK_OUT / IR |

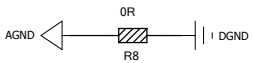
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HDMI CEC只能复用在GP_A0/A8/A22 TIMER3功能



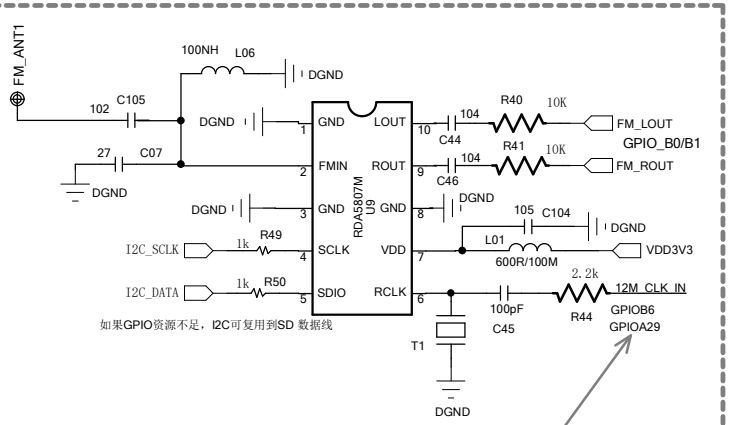

蓝牙天线必须采用山景推荐的型号(F型天线)
Layout时注意线宽及GND与天线之间的距离
BP1048B2芯片滤波电容需靠近芯片端放置且容量不能随意减小
详情见《硬件设计指南》

预留晶体负载电容位置，方便客户微调晶体时钟精度
晶体尽量考虑芯片管脚位置摆放
24M晶体选型要求：ESR电阻 ≤ 50 ohm
精度要求 ≤ 10ppm
Shunt Capacitance ≤ 5pF
晶体负载电容值默认选10pF

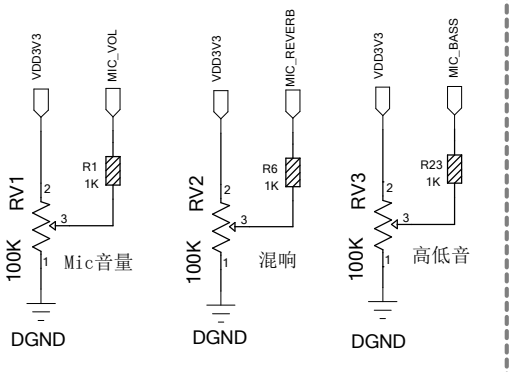
注意事项

1：AGND/DGND Pcb Layout需独立分地
单点连接至电源地

2：HDMI CEC口 可复用在TIMER功能的 GPIO口



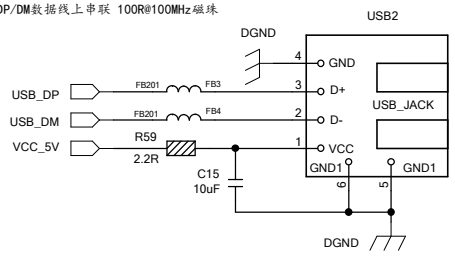

GPIOB6/A29 只能输出12M 时钟，不能输出32.768KHz时钟



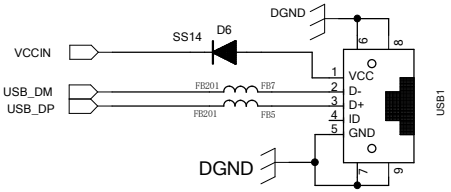

在线仿真调试接口



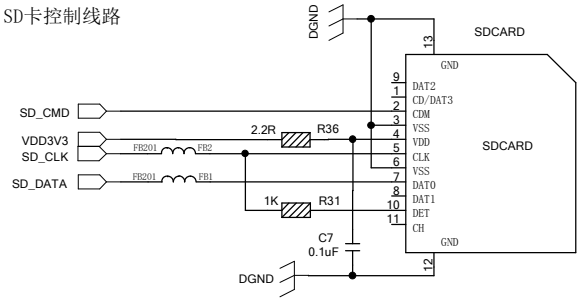

应EMC认证要求：需要在 DP/DM数据线上串联 100R@100MHz磁珠或者采用共模电感。



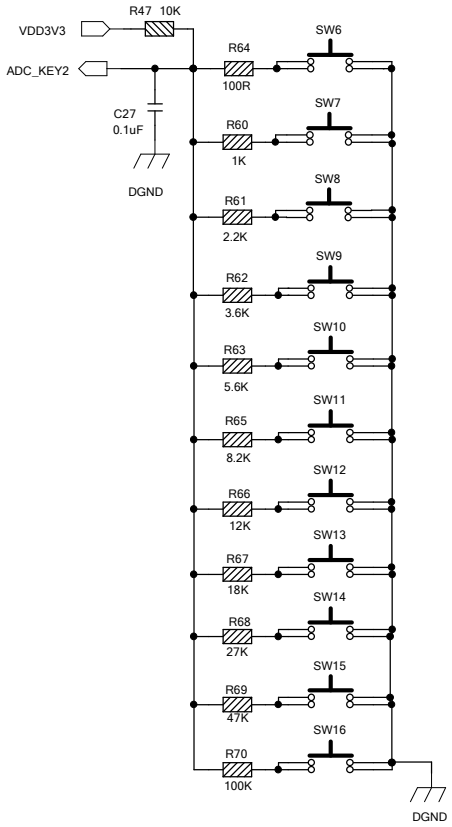

USB口同时复用调音接口（调音走HID协议）




SD卡控制线路

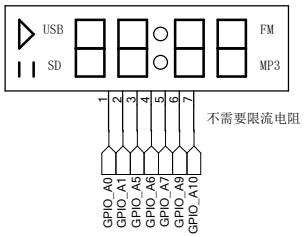



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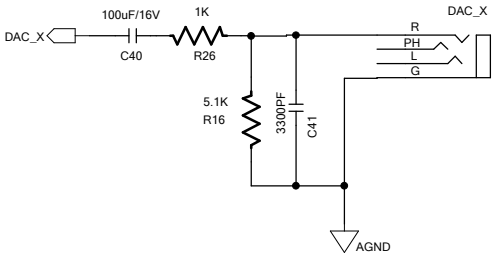

7PIN LED显示屏有指定选用的GPIO口

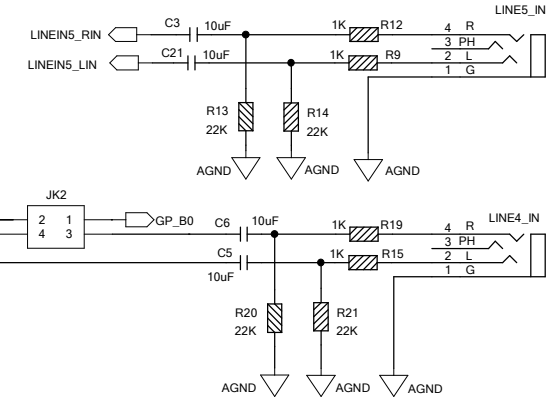
GPIOA0/A1、A5~A10 只能驱动8mA 档电流




7Pin LED屏

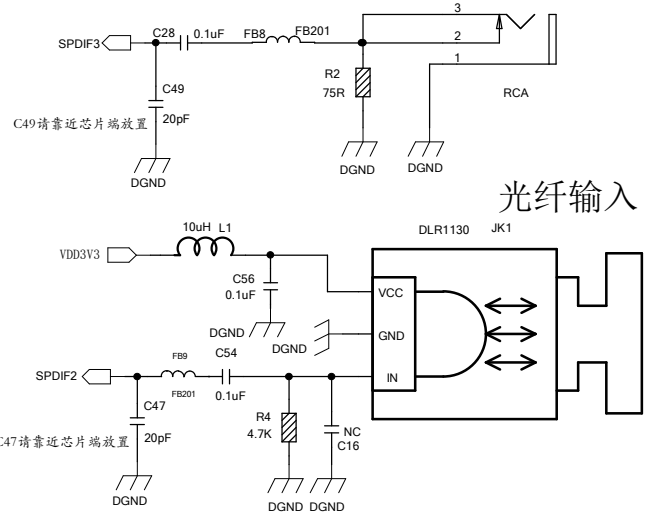
SUB 输出

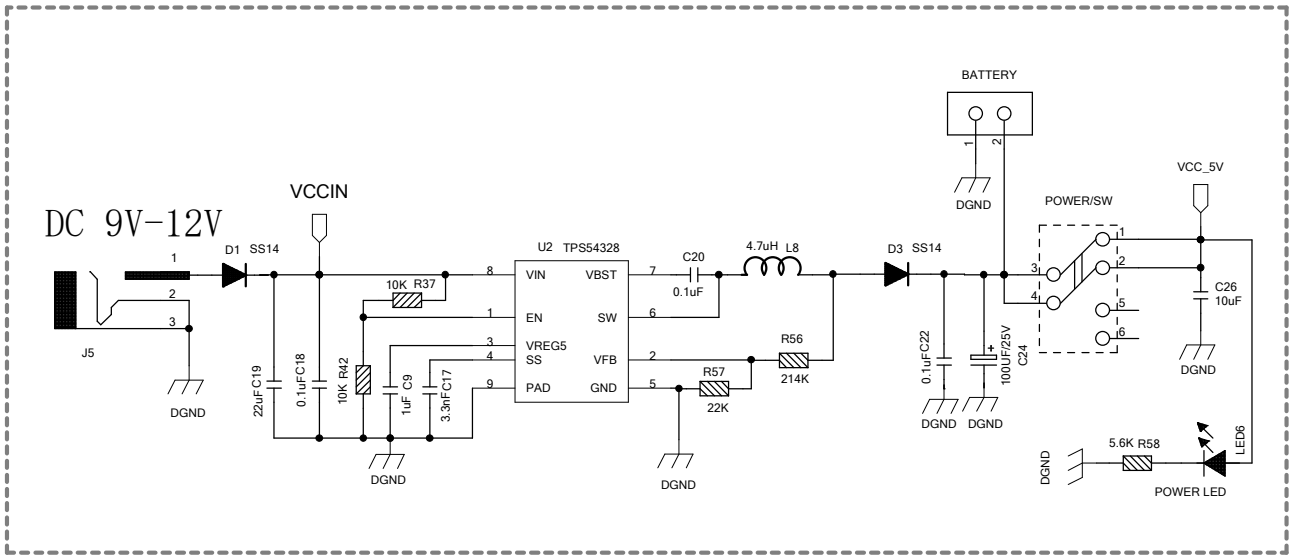



同轴输入

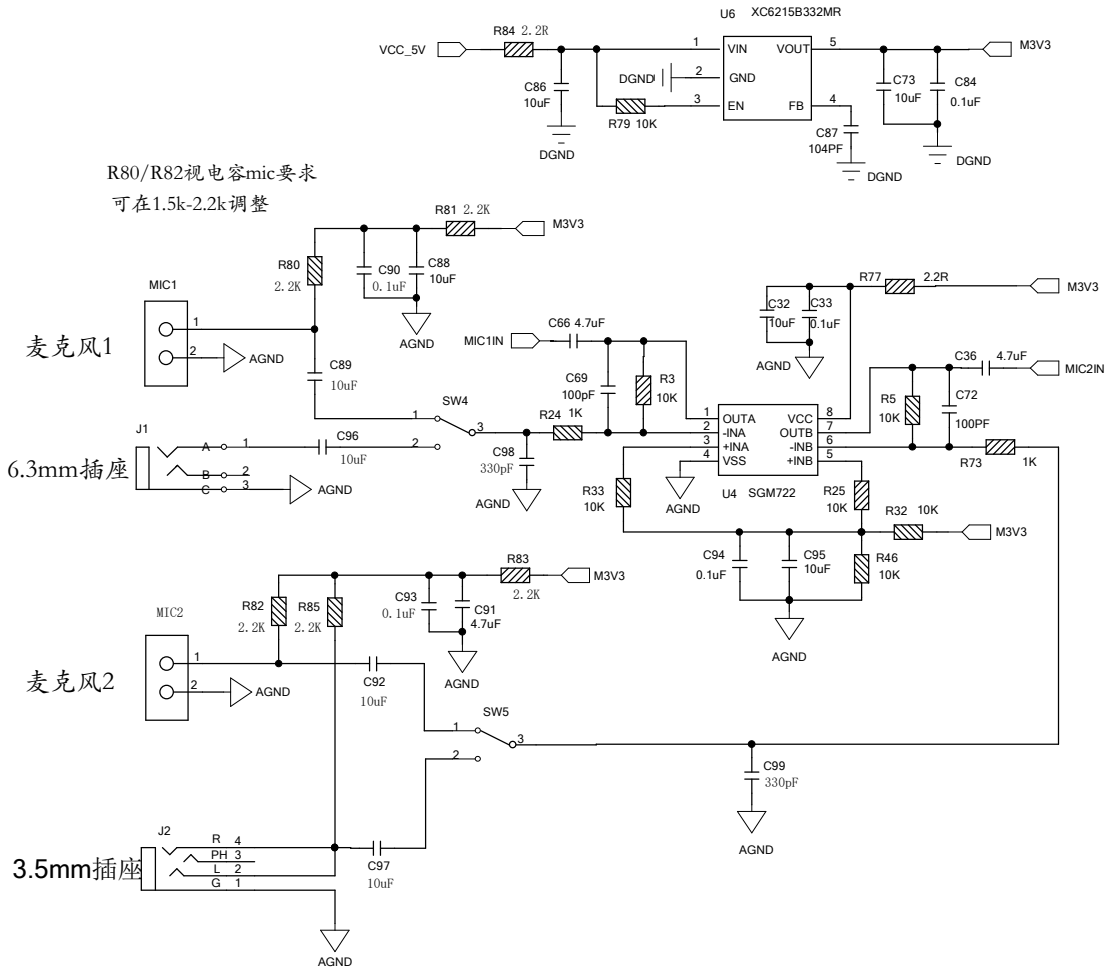
光纤输入

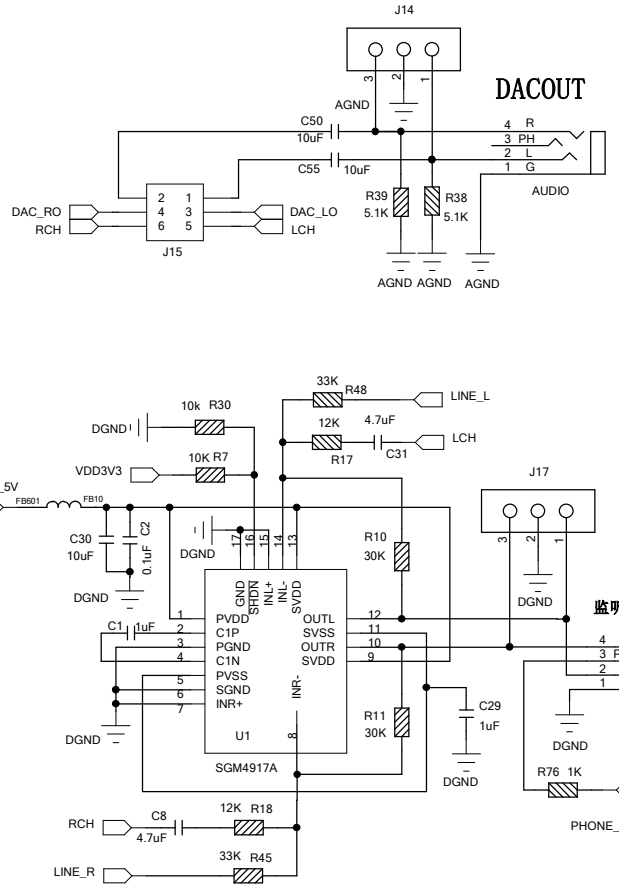


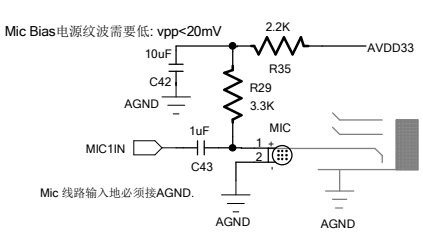




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# Karaoke应用前级放大器



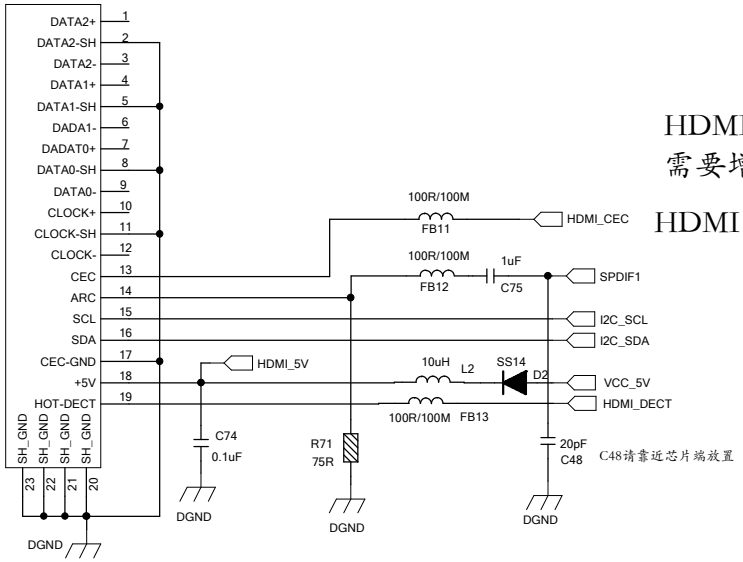




蓝牙通话应用的参考电路

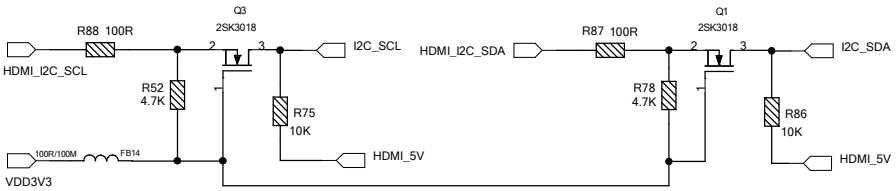
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## HDMI ARC




HDMI CEC 属于高频数据线路，对外有辐射干扰
需要增加铜箔地线对CEC线路进行屏蔽

HDMI CEC口只能复用在GP_A0/A8/A22 TIMER3功能



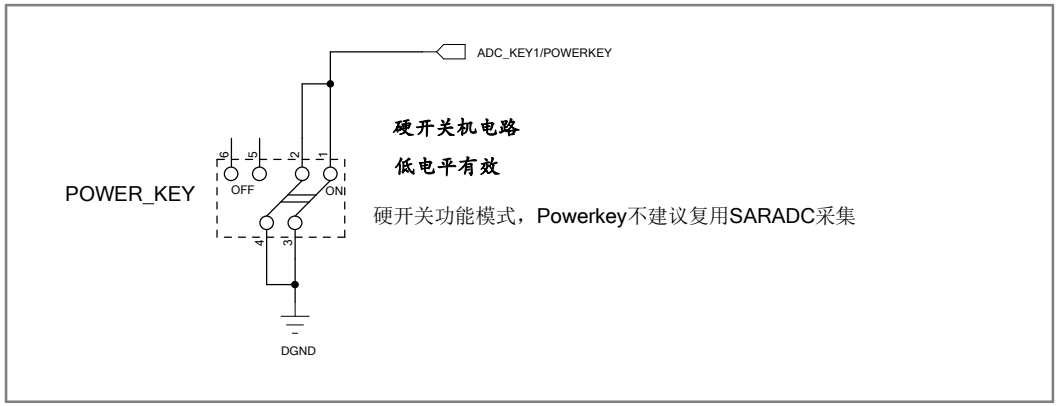

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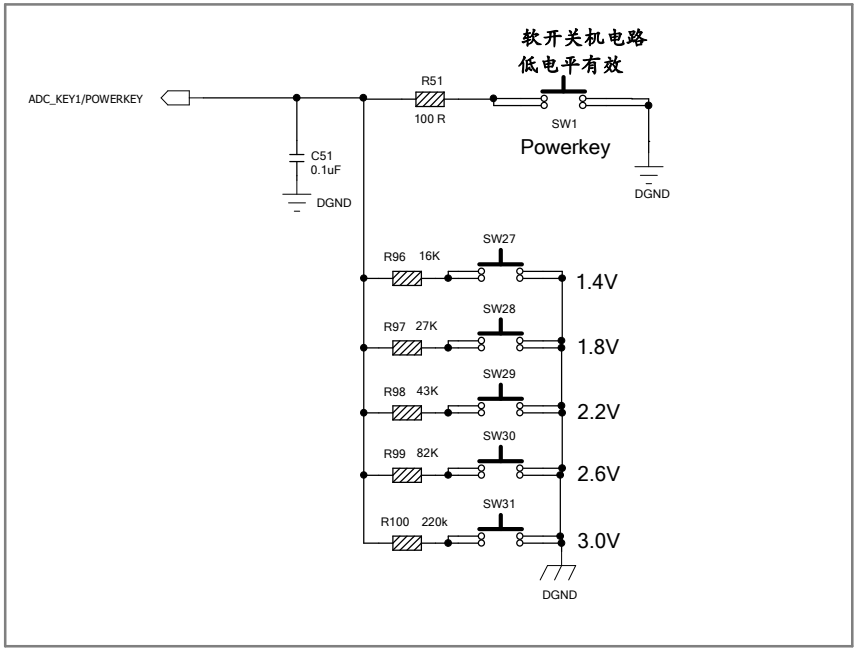
1、此POWERKEY管脚配置为POWERKEY功能并复用ADCKey时，
采集电压为1.2~3.3v，采样精度是7bit

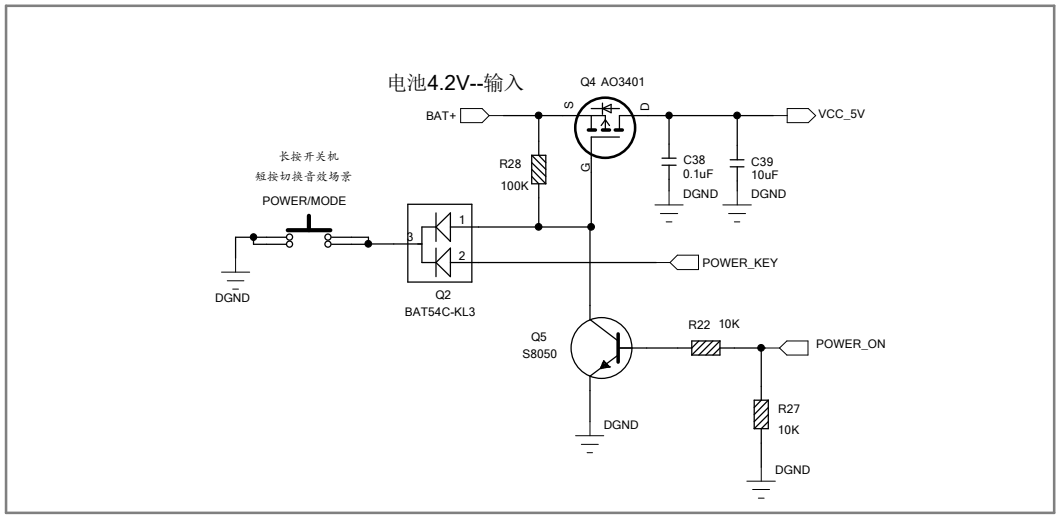
2、POWERKEY脚单独配置为SARADC，采集电压范围是：0~3.3v，最多可以支持11个按键采集
POWERKEY 管脚内部默认有上拉电阻值 22K / 高电平3.3V　IC硬件设计为Powerkey管脚的内部上拉电阻不能取消
外部不需要加上拉电阻

3、不使用 POWERKEY 功能时，请保持 POWERKEY 管脚悬空，
禁止将 POWERKEY 短接地或者连接到 5V

4、 POWERKEY 管脚做硬开关功能时，不能复用做SARADC 采集电压




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